

## CHAPTER IV

### SUCCESSIVE DIFFERENTIATION; LEIBNIZ THEOREM

1. Given  $y = x^n$ . Find  $y_n$ .

**Solution:**  $y_1 = nx^{n-1}$ ,  $y_2 = n(n-1)x^{n-2}$ ,  $y_3 = n(n-1)(n-2)x^{n-3}$ ,  
 $y_n = n(n-1)(n-2)\cdots 3.2.1x^{n-n} = n!$

2. Given  $y = (ax+b)^m$ . Find  $y_n$ .

**Solution:**  $y_1 = m(ax+b)^{m-1}a$ ,  $y_2 = m(m-1)(ax+b)^{m-2}a^2$ ,  $y_3 = m(m-1)(m-2)(ax+b)^{m-3}a^3$ ,  
 $y_n = m(m-1)(m-2)\cdots(m-n+1)(ax+b)^{m-n}a^n$

3. Given  $y = \log_e(ax+b)$ . Find  $y_n$ .

**Solution:**  $y_1 = \frac{1}{ax+b}a = (ax+b)^{-1}a$ ,  $y_2 = (-1)(ax+b)^{-2}a^2$ ,  
 $y_3 = (-1)(-2)(ax+b)^{-3}a^3 = (-1)^2 2!(ax+b)^{-3}a^3 = (-1)^{3-1} (3-1)!(ax+b)^{-3}a^3$   
 $y_n = (-1)^{n-1} (n-1)!(ax+b)^{-n}a^n$

4. Given  $y = \sin(ax+b)$ . Find  $y_n$ .

**Solution:**  $y_1 = a \cos(ax+b) = a \sin(\frac{\pi}{2} + ax+b)$   
 $y_2 = a^2 \cos(\frac{\pi}{2} + ax+b) = a^2 \sin(\frac{\pi}{2} + \frac{\pi}{2} + ax+b) = a^2 \sin(2\frac{\pi}{2} + ax+b)$   
 $y_3 = a^3 \cos(2\frac{\pi}{2} + ax+b) = a^3 \sin(3\frac{\pi}{2} + ax+b)$   
 $y_n = a^n \sin(n\frac{\pi}{2} + ax+b)$

#### Exercises

Find the n-th derivative of the following functions:

1.  $y = (ax+b)^{-m}$       2.  $y = \frac{1}{x+a}$       3.  $y = e^{ax}$       4.  $y = \cos(ax+b)$

#### Application of partial fraction for determining the n-th differentiation

##### Examples

1. Given  $y = \frac{2x+3}{x^2+3x+2}$ . Find  $y_n$ .

**Solution:** We can write  $y = \frac{2x+3}{x^2+3x+2} = \frac{2x+3}{(x+1)(x+2)} = \frac{1}{x+1} + \frac{1}{x+2} = (x+1)^{-1} + (x+2)^{-1}$

$y_1 = (-1)(x+1)^{-2} + (-1)(x+2)^{-2}$   
 $y_2 = (-1)(-2)(x+1)^{-3} + (-1)(-2)(x+2)^{-3} = (-1)^2 2!(x+1)^{-(2+1)} + (-1)^2 2!(x+2)^{-(2+1)}$   
 $y_3 = (-1)^3 3!(x+1)^{-(3+1)} + (-1)^3 3!(x+2)^{-(3+1)}$

$$y_n = (-1)^n n! (x+1)^{-(n+1)} + (-1)^n n! (x+2)^{-(n+1)} = (-1)^n n! [(x+1)^{-(n+1)} + (x+2)^{-(n+1)}]$$

### De Moivre's Theorem

[Abraham De Moivre (1667-1754), a French Mathematician]

For all real values of  $n$ ,  $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

1. Given  $y = \frac{1}{x^2 + a^2}$ . Find  $y_n$ .

**Solution:** We can write  $y = \frac{1}{x^2 + a^2} = \frac{1}{x^2 - i^2 a^2} = \frac{1}{(x - ia)(x + ia)} = \frac{1}{2ia} \left( \frac{1}{x - ia} - \frac{1}{x + ia} \right)$

$$= \frac{1}{2ia} [(x - ia)^{-1} - (x + ia)^{-1}]$$

$$y_1 = \frac{1}{2ia} [(-1)(x - ia)^{-2} - (-1)(x + ia)^{-2}]$$

$$y_2 = \frac{1}{2ia} [(-1)(-2)(x - ia)^{-3} - (-1)(-2)(x + ia)^{-3}]$$

$$= \frac{1}{2ia} [(-1)^2 2! (x+1)^{-(2+1)} + (-1)^2 2! (x+2)^{-(2+1)}]$$

$$y_3 = \frac{1}{2ia} [(-1)^3 3! (x+1)^{-(3+1)} + (-1)^3 3! (x+2)^{-(3+1)}]$$

$$y_n = \frac{1}{2ia} (-1)^n n! [(x - ia)^{-(n+1)} - (x + ia)^{-(n+1)}]$$

Let  $x = r \cos \theta$  and  $a = r \sin \theta$ .

Then  $x^2 + a^2 = r^2 (\cos^2 \theta + \sin^2 \theta) = r^2$  and  $\theta = \tan^{-1} \frac{a}{x}$  and  $r = \frac{a}{\sin \theta}$ .

Therefore,  $y_n = \frac{1}{2ia} (-1)^n n! [(r \cos \theta - ir \sin \theta)^{-(n+1)} - (r \cos \theta + ir \sin \theta)^{-(n+1)}]$

$$y_n = \frac{(-1)^n n!}{2iar^{n+1}} [\{\cos(n+1)\theta + i \sin(n+1)\theta\} - \{\cos(n+1)\theta - i \sin(n+1)\theta\}]$$

$$y_n = \frac{(-1)^n n!}{2iar^{n+1}} [2i \sin(n+1)\theta] = \frac{(-1)^n n!}{aa^{n+1}} [\sin^{n+1} \theta \sin(n+1)\theta]$$

Hence,  $y_n = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta$ , where  $\theta = \tan^{-1} \frac{a}{x}$ .

### Exercises

Find the  $n$ -th derivative of the following functions:

1.  $y = \frac{1}{1 - 5x + 6x^2}$     2.  $y = \frac{1}{x^2 + 5x + 6} + \cos x \cos 2x$     3.  $y = \frac{x^2}{(x-1)^2(x+2)}$

4.  $y = \frac{1}{x^2 + 16}$

### Problems

1. Find the n-th derivative of  $y = \cos x \cos 2x \cos 3x$ .

**Solution:**  $y = \cos x \cos 2x \cos 3x = \frac{1}{2}(2 \cos 3x \cos 2x) \cos x = \frac{1}{2}(\cos 5x + \cos x) \cos x$

$$= \frac{1}{2}(\cos 5x \cos x) + \frac{1}{2} \cos^2 x = \frac{1}{4}(2 \cos 5x \cos x) + \frac{1}{4}(2 \cos^2 x)$$
$$= \frac{1}{4}(\cos 6x + \cos 4x) + \frac{1}{4}(1 + \cos 2x) = \frac{1}{4}(\cos 6x + \cos 4x + \cos 2x + 1)$$

So that  $y_1 = \frac{1}{4}(-6 \sin 6x - 4 \sin 4x - 2 \sin 2x)$

$$= \frac{1}{4}[6 \cos(\frac{\pi}{2} + 6x) + 4 \cos(\frac{\pi}{2} + 4x) + 2 \cos(\frac{\pi}{2} + 2x)]$$

$$y_2 = \frac{1}{4}[-6^2 \sin(\frac{\pi}{2} + 6x) - 4^2 \sin(\frac{\pi}{2} + 4x) - 2^2 \sin(\frac{\pi}{2} + 2x)]$$

$$= \frac{1}{4}[6^2 \cos(2\frac{\pi}{2} + 6x) + 4^2 \cos(2\frac{\pi}{2} + 4x) + 2^2 \cos(2\frac{\pi}{2} + 2x)]$$

$$y_n = \frac{1}{4}[6^n \cos(n\frac{\pi}{2} + 6x) + 4^n \cos(n\frac{\pi}{2} + 4x) + 2^n \cos(n\frac{\pi}{2} + 2x)]$$

2. Find the n-th derivative of  $y = e^{ax} \sin bx$ .

**Solution:** Given  $y = e^{ax} \sin bx$ . So that  $y_1 = ae^{ax} \sin bx + be^{ax} \cos bx$ .

Let  $a = r \cos \theta$  and  $b = r \sin \theta$ . Then  $a^2 + b^2 = r^2(\cos^2 \theta + \sin^2 \theta) = r^2$  and  $\theta = \tan^{-1} \frac{b}{a}$ .

Therefore,  $y_1 = re^{ax} \sin bx \cos \theta + re^{ax} \cos bx \sin \theta = re^{ax} [\sin(bx + \theta)]$

$$y_2 = r^2 e^{ax} [\sin(bx + 2\theta)], \quad y_3 = r^3 e^{ax} [\sin(bx + 3\theta)].$$

Hence,  $y_n = r^n e^{ax} [\sin(bx + n\theta)] = (a^2 + b^2)^{\frac{n}{2}} e^{ax} [\sin(bx + n \tan^{-1} \frac{b}{a})]$ .

3. Given  $y = \cos^4 x$ . Find  $y_n$ .

**Solution:** Let  $z = \cos x + i \sin x$  so that  $z^{-1} = \frac{1}{z} = \cos x - i \sin x$ .

Therefore,  $z + \frac{1}{z} = 2 \cos x$  and so  $z^n + \frac{1}{z^n} = 2 \cos nx$

Now  $2^4 \cos^4 x = (z + \frac{1}{z})^4 = z^4 + {}^4C_1 z^{4-1} (\frac{1}{z})^1 + {}^4C_2 z^{4-2} (\frac{1}{z})^2 + {}^4C_3 z^{4-3} (\frac{1}{z})^3 + (\frac{1}{z})^4$

$$= z^4 + 4z^2 + 6 + 4(\frac{1}{z})^2 + (\frac{1}{z})^4 = (z^4 + \frac{1}{z^4}) + 4(z^2 + \frac{1}{z^2}) + 6$$

$$= 2 \cos 4x + 4(2 \cos 2x) + 6$$

$$\Rightarrow \cos^4 x = \frac{1}{8}[\cos 4x + 4 \cos 2x + 3] = y$$

$$\text{So, } y_1 = \frac{1}{8}[-4\sin 4x - 4.2\sin 2x] = \frac{1}{8}[4\cos(\frac{\pi}{2} + 4x) + 4.2\cos(\frac{\pi}{2} + 2x)]$$

$$y_2 = \frac{1}{8}[-4^2\sin(\frac{\pi}{2} + 4x) - 4.2^2\sin(\frac{\pi}{2} + 2x)] = \frac{1}{8}[4^2\cos(2\frac{\pi}{2} + 4x) + 4.2^2\cos(2\frac{\pi}{2} + 2x)]$$

$$y_n = \frac{1}{8}[4^n\cos(n\frac{\pi}{2} + 4x) + 4.2^n\cos(n\frac{\pi}{2} + 2x)] = 2^{2n-3}\cos(n\frac{\pi}{2} + 4x) + 2^{n-1}\cos(n\frac{\pi}{2} + 2x)$$

4. Given  $y = \cos^5 x$ . Find  $y_n$ .

**Solution:** Let  $z = \cos x + i \sin x$  so that  $z^{-1} = \frac{1}{z} = \cos x - i \sin x$ .

Therefore,  $z + \frac{1}{z} = 2\cos x$  and so  $z^n + \frac{1}{z^n} = 2\cos nx$

$$\text{Now } 2^5 \cos^5 x = (z + \frac{1}{z})^5$$

$$= z^5 + {}^5C_1 z^{5-1}(\frac{1}{z})^1 + {}^5C_2 z^{5-2}(\frac{1}{z})^2 + {}^5C_3 z^{5-3}(\frac{1}{z})^3 + {}^5C_4 z^{5-4}(\frac{1}{z})^4 + (\frac{1}{z})^5$$

$$= z^5 + 5z^3 + 10z + 10(\frac{1}{z}) + 5(\frac{1}{z})^4 + (\frac{1}{z})^5$$

$$= (z^5 + \frac{1}{z^5}) + 5(z^3 + \frac{1}{z^3}) + 10(z + \frac{1}{z})$$

$$= 2\cos 5x + 5(2\cos 3x) + 10(2\cos x)$$

$$\Rightarrow \cos^5 x = \frac{1}{32}[\cos 5x + 5\cos 3x + 10\cos x] = y$$

$$\text{So, } y_1 = \frac{1}{32}[-5\sin 5x - 5.3\sin 3x - 10\sin x]$$

$$= \frac{1}{32}[5\cos(\frac{\pi}{2} + 5x) + 5.3\cos(\frac{\pi}{2} + 3x) + 10\cos(\frac{\pi}{2} + x)]$$

$$y_2 = \frac{1}{32}[-5^2\sin(\frac{\pi}{2} + 5x) - 5.3^2\sin(\frac{\pi}{2} + 3x) - 10\sin(\frac{\pi}{2} + x)]$$

$$= \frac{1}{32}[5^2\cos(2\frac{\pi}{2} + 5x) + 5.3^2\cos(2\frac{\pi}{2} + 3x) + 10\cos(2\frac{\pi}{2} + x)]$$

$$y_n = \frac{1}{32}[5^n\cos(n\frac{\pi}{2} + 5x) + 5.3^n\cos(n\frac{\pi}{2} + 3x) + 10\cos(n\frac{\pi}{2} + x)]$$

5. Given  $y = \sin^6 x$ . Find  $y_n$ .

**Solution:** Let  $z = \cos x + i \sin x$  so that  $z^{-1} = \frac{1}{z} = \cos x - i \sin x$ .

Therefore,  $z - \frac{1}{z} = 2i \sin x$  and so  $z^n + \frac{1}{z^n} = 2\cos nx$

$$\text{Now } 2^6 i^6 \sin^6 x = (z - \frac{1}{z})^6$$

$$\begin{aligned}
&= z^6 - {}^6C_1 z^{6-1} \left(\frac{1}{z}\right)^1 + {}^6C_2 z^{6-2} \left(\frac{1}{z}\right)^2 - {}^6C_3 z^{6-3} \left(\frac{1}{z}\right)^3 + {}^6C_4 z^{6-4} \left(\frac{1}{z}\right)^4 - {}^6C_5 z^{6-5} \left(\frac{1}{z}\right)^5 + \left(\frac{1}{z}\right)^6 \\
&= z^6 - 6z^4 + 15z^2 - 20 + 15\left(\frac{1}{z}\right)^2 - 6\left(\frac{1}{z}\right)^4 + \left(\frac{1}{z}\right)^6 \\
&= \left(z^6 + \frac{1}{z^6}\right) - 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) - 20 \\
&= 2 \cos 6x - 6(2 \cos 4x) + 15(2 \cos 2x) - 20 \\
\Rightarrow \sin^6 x &= -\frac{1}{32} [\cos 6x - 6 \cos 4x + 15 \cos 2x - 10] = y \\
\text{So, } y_1 &= -\frac{1}{32} [-6 \sin 6x - 6.4 \sin 4x - 15.2 \sin 2x] \\
&= -\frac{1}{32} [6 \cos\left(\frac{\pi}{2} + 6x\right) - 6.4 \cos\left(\frac{\pi}{2} + 4x\right) - 15.2 \cos\left(\frac{\pi}{2} + 2x\right)] \\
y_2 &= -\frac{1}{32} [-6^2 \sin\left(\frac{\pi}{2} + 6x\right) - 6.4^2 \sin\left(\frac{\pi}{2} + 4x\right) - 15.2^2 \sin\left(\frac{\pi}{2} + 2x\right)] \\
&= -\frac{1}{32} [6^2 \cos\left(2\frac{\pi}{2} + 6x\right) - 6.4^2 \cos\left(2\frac{\pi}{2} + 4x\right) + 15.2^2 \cos\left(2\frac{\pi}{2} + 2x\right)] \\
y_n &= -\frac{1}{32} [6^n \cos\left(n\frac{\pi}{2} + 6x\right) - 6.4^n \cos\left(n\frac{\pi}{2} + 4x\right) + 15.2^n \cos\left(n\frac{\pi}{2} + 2x\right)]
\end{aligned}$$

### Problems

1. Find  $y_2$ , where  $x = a(\cos \theta + \theta \sin \theta)$  and  $y = a(\sin \theta - \theta \cos \theta)$ .
2. If  $y = \sin(\sin x)$ , then prove that  $\frac{d^2 y}{dx^2} + \tan x \frac{dy}{dx} + y \cos x = 0$ .
3. If  $y = x \ln[(ax)^{-1} + a^{-1}]$ , then prove that  $x(x+1) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 1 = y$ .
4. If  $y = e^{-x} \sin x$ , then prove that  $\frac{d^4 y}{dx^4} + 4y = 0$ .

## Leibniz Theorem

**Statement:** If  $u(x)$  and  $v(x)$  are two functions of  $x$ , then the  $n$ -th derivative of their product is

denoted by  $(uv)_n$  or  $\frac{d^n}{dx^n}(uv)$  and is defined by

$$(uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \cdots + {}^nC_r u_{n-r} v_r + \cdots + uv_n$$

where the suffixes of  $u$  and  $v$  denote the differentiation of  $u$  and  $v$  with respect to  $x$ .

## Solved Problems

1. Given  $\cos^{-1}\left(\frac{y}{b}\right) = \ln\left(\frac{x}{m}\right)^m$ . Prove that  $x^2 y_{n+2} + (2n+1)xy_{n+1} + (m^2 + n^2)y_n = 0$ .

**Solution:** We have  $\cos^{-1}\left(\frac{y}{b}\right) = \ln\left(\frac{x}{m}\right)^m$ . We can write

$$\frac{y}{b} = \cos\left[\ln\left(\frac{x}{m}\right)^m\right] \Rightarrow \frac{y}{b} = \cos\left(m \ln \frac{x}{m}\right) \Rightarrow y = b \cos\left(m \ln \frac{x}{m}\right)$$

So that  $y_1 = -b \sin\left(m \ln \frac{x}{m}\right) \times m \times \frac{1}{x} = -\frac{bm}{x} \sin\left(m \ln \frac{x}{m}\right)$

$$\Rightarrow xy_1 = -bm \sin\left(m \ln \frac{x}{m}\right)$$

$$\Rightarrow xy_2 + y_1 = -bm \cos\left(m \ln \frac{x}{m}\right) \times m \times \frac{1}{x} = -\frac{m^2}{x} \times y$$

$$\Rightarrow x^2 y_2 + xy_1 + m^2 y = 0$$

Applying Leibniz's theorem, we get

$$x^2 y_{n+2} + ny_{n+1} \times 2x + \frac{n(n-1)}{2} y_n \times 2 + xy_{n+1} + ny_n + m^2 y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + 2nxy_{n+1} + (n^2 - n)y_n + xy_{n+1} + ny_n + m^2 y_n = 0$$

$$\text{Hence } x^2 y_{n+2} + (2n+1)xy_{n+1} + (m^2 + n^2)y_n = 0.$$

2. Given  $x = \sin\left(\frac{\ln y}{m}\right)$ . By applying Leibniz's theorem prove that

$$(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + m^2)y_n = 0.$$

**Solution:** We have  $x = \sin\left(\frac{\ln y}{m}\right)$ . We can write

$$\sin^{-1} x = \frac{1}{m} \ln y \Rightarrow \ln y = m \sin^{-1} x \quad (1)$$

Differentiating (1) with respect to  $x$ , we get

$$\frac{y_1}{y} = \frac{m}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2} y_1 = my \Rightarrow (1-x^2) y_1^2 = m^2 y^2 \quad (2)$$

Again, differentiating (2) with respect to  $x$ , we get

$$\begin{aligned} 2(1-x^2)y_1 y_2 + y_1^2(-2x) &= m^2(2yy_1) \\ \Rightarrow (1-x^2)y_2 - x y_1 - m^2 y &= 0 \end{aligned} \quad (3)$$

Applying Leibniz's theorem, we get

$$\begin{aligned} (1-x^2)y_{n+2} + ny_{n+1} \times (-2x) + \frac{n(n-1)}{2} y_n \times (-2) - (xy_{n+1} + ny_n) - m^2 y_n &= 0 \\ \Rightarrow (1-x^2)y_{n+2} - 2nxy_{n+1} - (n^2 - n + n + m^2)y_n - xy_{n+1} &= 0 \end{aligned}$$

Hence  $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + m^2)y_n = 0$ .

3. If  $y = (x^2 - 1)^n$ , then show that

$$(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0.$$

**Solution:** We have  $y = (x^2 - 1)^n$  (1)

Differentiating (1) with respect to  $x$ , we get

$$\begin{aligned} y_1 &= n(x^2 - 1)^{n-1}(2x) = \frac{2nx(x^2 - 1)^n}{x^2 - 1} \\ \Rightarrow (x^2 - 1)y_1 &= 2nx(x^2 - 1)^n = 2nxy, \text{ by (1)} \end{aligned} \quad (2)$$

Again, differentiating (2) with respect to  $x$ , we get

$$\begin{aligned} (x^2 - 1)y_2 + 2xy_1 &= 2ny + 2nxy_1 \\ \Rightarrow (x^2 - 1)y_2 + 2(1-n)xy_1 - 2ny &= 0 \end{aligned} \quad (3)$$

Applying Leibniz's theorem, we get

$$\begin{aligned} (x^2 - 1)y_{n+2} + ny_{n+1} \times (2x) + \frac{n(n-1)}{2} y_n \times (2) - 2(n-1)(xy_{n+1} + ny_n) - 2ny_n &= 0 \\ \Rightarrow (x^2 - 1)y_{n+2} + 2(n-n+1)xy_{n+1} + (n^2 - n - 2n^2 + 2n - 2n)y_n &= 0 \end{aligned}$$

Hence  $(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$ .

4. If  $y = [x + \sqrt{1+x^2}]^m$ , then show that

$$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$

**Solution:** We have  $y = [x + \sqrt{1+x^2}]^m$  (1)

Differentiating (1) with respect to  $x$ , we get

$$\begin{aligned} y_1 &= m[x + \sqrt{1+x^2}]^{m-1} \left[ 1 + \frac{2x}{2\sqrt{1+x^2}} \right] = \frac{m[x + \sqrt{1+x^2}]^m}{x + \sqrt{1+x^2}} \left( \frac{x + \sqrt{1+x^2}}{\sqrt{1+x^2}} \right) \\ \Rightarrow \sqrt{1+x^2} y_1 &= my \Rightarrow (1+x^2) y_1^2 = m^2 y^2, \text{ by (1)} \end{aligned} \quad (2)$$

Again, differentiating (2) with respect to  $x$ , we get

$$\begin{aligned} 2(1+x^2)y_1 y_2 + y_1^2(2x) &= m^2(2yy_1) \\ \Rightarrow (1+x^2)y_2 + xy_1 - m^2 y &= 0 \end{aligned} \quad (3)$$

Applying Leibniz's theorem, we get

$$(1+x^2)y_{n+2} + ny_{n+1} \times (2x) + \frac{n(n-1)}{2} y_n \times (2) + (xy_{n+1} + ny_n) - m^2 y_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - n + n - m^2)y_n = 0$$

$$\text{Hence } (1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$

### Exercises

1. If  $y = e^{m \cos^{-1} x}$ , then by applying Leibniz's theorem show that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2 + n^2)y_n = 0.$$

2. If  $x = \sin\left(\frac{1}{m} \sin^{-1} y\right)$ , then by applying Leibniz's theorem prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0.$$

3. If  $y = \sin(m \sin^{-1} x)$ , then by applying Leibniz's theorem show that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0.$$

4. If  $\cos^{-1}\left(\frac{y}{b}\right) = \ln\left(\frac{x}{n}\right)^n$  then prove that

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + 2n^2 y_n = 0.$$

5. If  $y = A \cos\{m \sin^{-1}(ax+b)\}$ , then show that

$$\{1 - (ax+b)^2\}y_{n+2} - (2n+1)a(ax+b)y_{n+1} + (m^2 - n^2)a^2 y_n = 0.$$

6. If  $y^{\frac{1}{m}} + y^{-\frac{1}{m}} = 2x$ , then show that

$$(x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$

7. If  $y = a \cos(\ln x) + b \sin(\ln x)$ , then by applying Leibniz's theorem show that

$$(i) \quad x^2 y_2 + x y_1 + y = 0.$$

$$(ii) \quad x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2 + 1)y_n = 0.$$